

# An inter-city, cross-species optical clock comparison at the $5 \times 10^{-18}$ level over a 1350 km fibre link

XXXXX

## Abstract

A redefinition of the SI second requires optical clocks in different laboratories to measure the same frequency ratio independently and to agree within  $5 \times 10^{-18}$ . Several clocks now report self-evaluated uncertainties at this level, but no comparison between clocks in different cities has yet verified the requirement. Here we compare a ytterbium lattice clock in Wuhan with a strontium lattice clock in Shanghai, about 700 km apart, through a 1350 km phase-stabilised fibre link, and determine the Yb/Sr frequency ratio from 58 days of coherent operation (986 416 retained seconds across 16 segments). Carrying the time-varying gravitational-redshift term as an uncertainty item, we obtain  $\text{Yb/Sr} = 1.207\,507\,039\,343\,337\,720\,6(21)$ , with a statistical uncertainty of  $0.89 \times 10^{-18}$  and a total standard uncertainty of  $2.1 \times 10^{-18}$ . The central value differs from the NIST–JILA result by  $-2.44 \times 10^{-18}$  and from the European-network result by  $+2.56 \times 10^{-18}$ , both within the redefinition threshold. To our knowledge, this is the first reported inter-city, cross-species frequency-ratio measurement meeting the uncertainty requirement of the roadmap towards the SI-second redefinition.

## 1 Introduction

Optical lattice and single-ion clocks now reach fractional systematic uncertainties at or below  $1 \times 10^{-18}$ <sup>1,2</sup>, and two-clock instabilities approach  $10^{-18}$  within an hour<sup>3</sup>. Their value compounds when they are compared. Comparing independent clocks verifies accuracy budgets that no single laboratory can test, because the environmental and geodetic assumptions entering a self-evaluation are shared within one site; linking clocks at different locations into a cooperative *network* is therefore how optical clocks become verifiable metrological infrastructure<sup>4,5</sup>. The idea has moved from proposal to practice: fibre links with added noise below the clocks themselves now span thousands of kilometres<sup>6,7</sup>, and multi-clock campaigns have measured tens of ratios among clocks in six countries<sup>8</sup> and across a continental network<sup>9</sup>.

The most immediate purpose of such a network is the redefinition of the SI second. The roadmap towards the redefinition requires, among its mandatory criteria, that frequency ratios be measured independently in different laboratories with an agreement better than  $5 \times 10^{-18}$ , and that individual ratios be repeated by different institutes<sup>10</sup>; the requirement was restated in the 2025 CCTF progress reports<sup>11,12</sup>. The present status is asymmetric. Ratios at or below the threshold have been demonstrated only between clocks on the same campus, most recently over a 3.6 km stabilised link with uncertainties at or below  $3.2 \times 10^{-18}$ <sup>13</sup>; comparisons across independent institutions have reached  $7.7 \times 10^{-18}$  at best<sup>9</sup>. No frequency-ratio measurement between clocks in different cities has yet met the requirement<sup>14</sup>.

Closing that gap matters beyond the definition itself. Chronometrically levelled clocks map the gravitational potential of the Earth at centimetre resolution<sup>15–17</sup>; clock networks have been proposed and used as sensors for dark matter<sup>18,19</sup>; and intercontinental clock comparisons through very long baseline interferometry (VLBI) have linked clocks 9000 km apart in Italy and Japan<sup>20</sup>. All of these applications assume that remote clocks agree at a level that, until now, has only been demonstrated locally.

Here we report an inter-city, cross-species frequency-ratio measurement between an IAPMST ytterbium lattice clock in Wuhan and a USTC strontium lattice clock in Shanghai, about 700 km



Figure 1: The two-clock network. An IAPMST ytterbium clock in Wuhan and a USTC strontium clock in Shanghai are connected by a phase-stabilised 1550 nm fibre link. Relay stations along the route carry erbium-doped fibre amplifiers and fibre-noise-cancellation loops.

apart, connected by a 1350 km phase-stabilised fibre link and operated coherently for 58 days in three stages. From 16 retained data segments (986 416 seconds), we determine the Yb/Sr ratio to a total standard uncertainty of  $2.1 \times 10^{-18}$ , meeting the uncertainty requirement of the redefinition roadmap; the time-varying gravitational-redshift term is carried as an uncertainty item rather than applied as a correction. The result establishes the two sites as a prototype node pair for a distributed optical-clock network, and completes, at inter-city scale, the class of ratio measurements that the redefinition process depends on.

## 2 Results

### 2.1 Network and observation campaign

Figure 1 shows the two-clock network. The Yb and Sr clocks are separated by 700 km and connected through a 1350 km phase-stabilised fibre link operated at 1550 nm; the loop-back topology and the relay stations keep the link's added instability below the clock noise for averaging times beyond 100 s. Over two months (58 days, 3 stages) the two clocks and their transfer chains ran synchronously for a total of 1 008 912 valid seconds, recorded as 17 jump-free segments. After excluding one segment whose Sr systematic parameter is pending verification (Section 4.2), the primary data set contains 16 segments and 986 416 seconds (274 effective hours). Figure 2a places the segments on the campaign timeline.

### 2.2 Clock-ratio determination

We determine the Yb/Sr frequency ratio from the 16 segments. Each segment yields a ratio estimate with its own statistical uncertainty estimated from the Allan deviation of the beat (Section 4.3). The segment estimates scatter more than their individual uncertainties allow: the fixed-effect weighted least-squares fit gives  $\chi^2_{\text{red}} = 4.017$  for 15 degrees of freedom ( $p = 2.3 \times 10^{-7}$ ), and 5 of the 16 segments have stability fits that fall outside the white-frequency-noise band (groups 2, 6, 7, 12, 16). We therefore adopt the Bayesian random-effect estimate, which treats the excess scatter as a free parameter and propagates its posterior uncertainty, while the fixed-effect WLS and Mandel–Paule estimates are kept as cross-checks.

The resulting Yb/Sr ratio is

$$\text{Yb/Sr} = 1.207\,507\,039\,343\,337\,720\,6(21), \quad (1)$$

with a statistical uncertainty of  $0.89 \times 10^{-18}$  and a total standard uncertainty of  $2.1 \times 10^{-18}$  (Section 2.5). The precision-weighted cross-checks agree within their uncertainties: the WLS centre

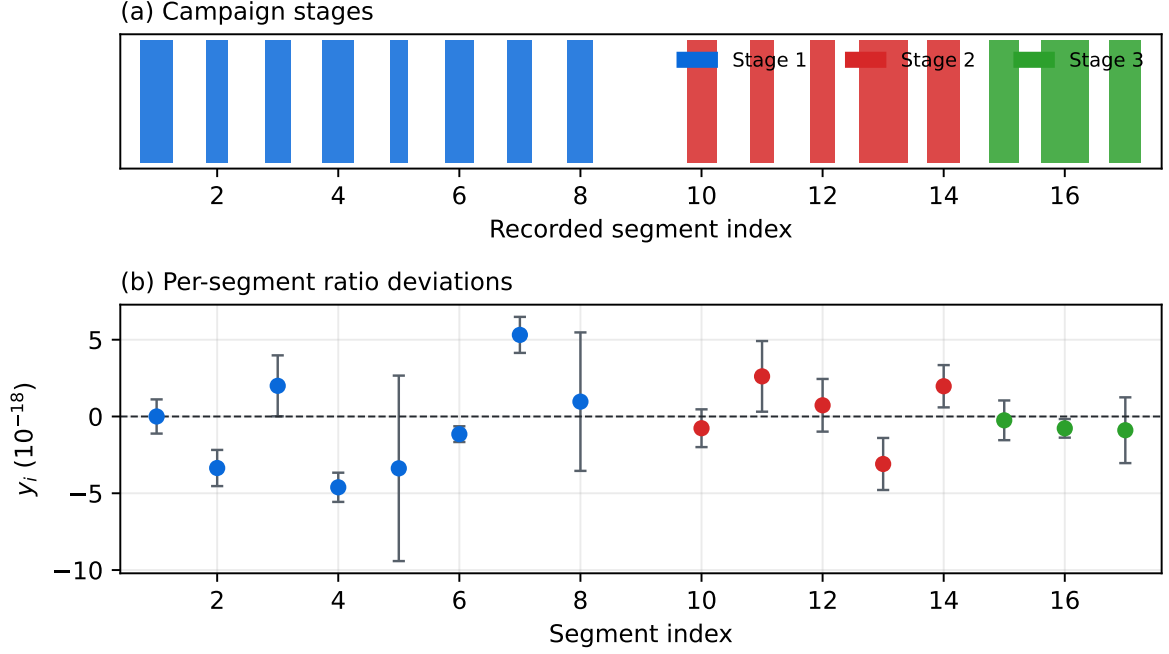


Figure 2: Campaign and per-segment data. (a) Recorded segments coloured by measurement stage; bar widths scale with the square root of the retained sample count. (b) Per-segment ratio deviations  $y_i$  with their statistical uncertainties; the dashed line marks zero deviation.

is 1.207 507 039 343 337 720 0 and the Mandel–Paule centre is 1.207 507 039 343 337 720 6 (rounded at the 19th decimal), both consistent with the adopted value at the  $10^{-18}$  level. Figure 2b shows the per-segment deviations; the segment-to-segment scatter is dominated by the statistical noise of the individual segments and by the excess scatter captured by the random-effect term.

### 2.3 Comparison with other measurements

The value reported here agrees with the two existing measurements of the Yb/Sr ratio at the level set by the redefinition roadmap (Figure 3). Relative to the NIST–JILA value 1.207 507 039 343 337 723 0(37)<sup>13</sup>, our central value differs by  $-2.44 \times 10^{-18}$ ; relative to the European-network value 1.207 507 039 343 337 718(32)<sup>9</sup>, it differs by  $+2.56 \times 10^{-18}$ . Both differences lie within the  $5 \times 10^{-18}$  agreement threshold of the roadmap’s frequency-ratio criterion<sup>10,12</sup>. To our knowledge, this is the first reported inter-city (100-km-scale), cross-species frequency-ratio measurement meeting that uncertainty requirement (as of *October 2026*); existing ratios at or below the threshold were measured on a single campus<sup>13</sup>, while inter-institute comparisons through fibre networks have so far reached  $7.7 \times 10^{-18}$ <sup>9</sup>.

### 2.4 Segment-9 sensitivity check

The excluded segment (section 4.2) carries 22496 seconds. Reinstating it changes the WLS centre by  $+0.30 \times 10^{-18}$  and raises the reduced chi-square of the 17-segment combination to 5.24, reflecting the segment’s excess scatter; the adopted random-effect treatment is insensitive to this at the reported precision, and the 16-segment result remains the primary value.

### 2.5 Uncertainty budget

Table 1 collects the components of the total standard uncertainty. The statistical term dominates the budget, followed by the Yb and Sr systematic evaluations, the static-potential term from the

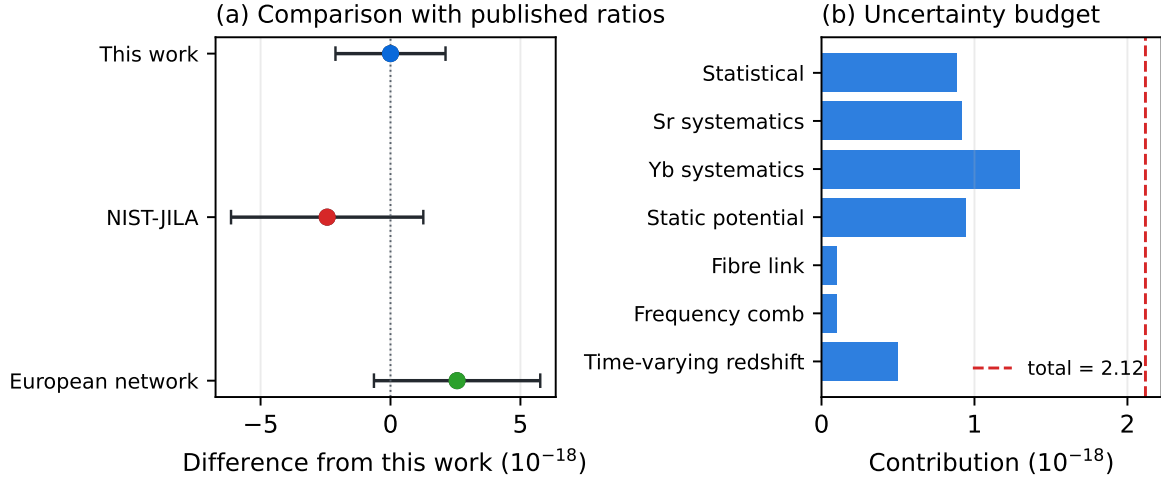


Figure 3: Comparison and budget. (a) Differences between this work and the published Yb/Sr ratios, with their quoted uncertainties. (b) Components of the total standard uncertainty; the dashed line marks the quadrature total.

Table 1: Uncertainty budget of the Yb/Sr frequency-ratio measurement. The time-varying gravitational-redshift term is carried as an uncertainty item and is not applied as a correction to the central value. Components are treated as independent.

| Component                            | Contribution ( $10^{-18}$ ) |
|--------------------------------------|-----------------------------|
| Statistical (Bayesian random effect) | 0.885                       |
| Sr systematics                       | 0.92                        |
| Yb systematics                       | 1.3                         |
| Static gravitational potential       | 0.946                       |
| Fibre link                           | 0.1                         |
| Frequency comb                       | 0.1                         |
| Time-varying redshift                | 0.5                         |
| Total (quadrature)                   | 2.12                        |

levelling, and the time-varying-redshift term carried as an uncertainty item; the fibre-link and comb terms are negligible at this precision. The total standard uncertainty is  $2.1 \times 10^{-18}$ .

### 3 Discussion

The result reported here closes, at inter-city scale, the class of frequency-ratio measurements that the SI-second redefinition process depends on. Meeting the roadmap’s frequency-ratio requirement has so far required clocks on a shared campus, where a single stabilised link serves both systems; here the two clocks sit in different cities, in different institutions, and use different atomic species, and the measurement still reaches the threshold, because the fibre link and the two combs contribute well below the clocks themselves. The bottleneck for reaching the redefinition threshold is therefore no longer the transfer infrastructure but the systematic evaluations of the clocks and the geodetic determination of their potential difference.

The campaign also demonstrates the operational character that a future optical-clock network will need. Over the two-month campaign the two clocks were restarted, the transfer chain re-locked, and the beat record interrupted many times; the segment-level ratio estimates show no

evidence of a systematic offset between the three measuring stages, so the reproducibility of the comparison does not depend on continuous operation of either system. This is the practical counterpart of the coordinated multi-clock campaigns<sup>8;9</sup>: the network tolerates interruption, provided each retained window is self-consistently characterised.

Several limitations bound the present result. First, the per-segment statistical uncertainties from the Allan-deviation extrapolation are the dominant budget term, and five of the sixteen segments carry model-limited fits; the random-effect treatment accounts for their excess scatter at the level of the reported uncertainty, but a longer record would reduce the statistical term. Second, the systematic evaluations of the two clocks are those published by the laboratories; independent verification of those budgets, as opposed to the transfer, is not attempted here. Third, the geopotential difference is taken from a first-order levelling campaign; an alternative value exists in the experiment records and is carried as an external item pending reconciliation, as is the physical-model provenance of the supplied tide series and the exact accounting of the full sample totals. Finally, the present campaign meets the uncertainty requirement of the roadmap but not the full criterion, which additionally requires each ratio to be measured at least twice by different institutes and repeated over an extended period.

Looking ahead, the result establishes this link as a prototype node pair of a distributed optical-clock network: the same infrastructure can be extended to further nodes, and the technique demonstrated here – clocks at different sites agreeing at the redefinition threshold – is the operation that both the redefinition and post-redefinition applications require. Realising those applications over larger distances will mean combining fibre networks of the kind used here<sup>6;7</sup> with intercontinental techniques such as VLBI<sup>20</sup>, and will put the same demands on geodetic infrastructure that this comparison faced<sup>15;17</sup>.

## 4 Methods

### 4.1 Experimental setup and frequency transfer

The comparison links two optical lattice clocks at different institutions: an ytterbium (Yb) clock at IAPMST in Wuhan and a strontium (Sr) clock at USTC in Shanghai, about 700 km apart. The two sites are connected by a phase-stabilised fibre link of 1 350 km single-pass length (2 700 km loop-back), operated at 1550 nm. The route is divided into spans of tens of kilometres, each terminated by a relay station that carries a narrow-linewidth laser together with an erbium-doped fibre amplifier and a fibre-noise-cancellation loop; the added instability of the link is kept below the clock noise level for averaging times beyond 100 s.

Both ends use independent optical frequency combs to bridge the spectral gap between the clock transitions and the telecom carrier. On the Shanghai side the Sr 698 nm transition is transferred via a 1397 nm continuous laser to the 1550 nm carrier; on the Wuhan side the received carrier is linked to the Yb 578 nm transition through the 1156 nm light. All local frequency references and phase-locked loops are derived from the comb repetition rate  $f_{\text{rep}} = 200$  MHz. The observable is the out-of-loop beat between the received carrier and the local Yb-referenced comb, converted to fractional units with the transfer frequency  $F_{1550} = 193.3992$  THz (see Appendix A).

### 4.2 Observation campaign and per-segment data

The campaign spanned 58 days, from 29 June to 26 August 2026, in 3 measurement stages. Synchronous-operation windows were first defined from the lock status of the two clocks and of the beat signals; within each window the longest continuous valid record was retained. The selection used lock status only and did not refer to the resulting ratio deviations. In total 17 jump-free segments were recorded, carrying 1 008 912 valid seconds.

The ninth recorded segment is excluded from the primary analysis. Its Sr systematic frequency-shift parameter sits between the values of the two operating periods and its provenance has not

been confirmed by the experiment team; the segment is therefore held pending verification. The exclusion is based on parameter provenance, not on the resulting ratio deviation, and its effect is quantified in the sensitivity check (Section 2.4). The primary data set thus comprises 16 segments and 986 416 valid seconds (274 effective hours), i.e. 22496 seconds fewer than the full record.

### 4.3 Clock-ratio determination and statistical model

Each retained segment yields a clock-ratio estimate  $R_i$  from the mean of its retained beat samples, inverted through the fixed frequency-chain relation (Appendix A) and corrected for the static gravitational redshift  $\delta_g = -3.116 \times 10^{-15}$  derived from the levelled geopotential difference (Section 4.4). The per-segment statistical uncertainty  $u_i$  is obtained from the overlapping Allan deviation of the beat fractional frequency: a  $\sigma_y \propto \tau^{-1/2}$  fit over the range  $128 \text{ s} \leq \tau \leq 0.25 T_i$ , extrapolated to the segment duration  $T_i$ .

The per-segment uncertainties do not fully explain the observed scatter between segments, so the ratio is combined with four estimators: fixed-effect weighted least squares (WLS, weights  $1/u_i^2$ ), Birge scaling, the Mandel–Paule random-effect model, and a Bayesian random-effect model in which the excess scatter  $\xi$  is a free parameter whose posterior uncertainty is propagated into the final result. The Bayesian random-effect centre is adopted as the primary statistic; the others are reported as cross-checks. Five of the 16 segments (groups 2, 6, 7, 12, 16) have stability fits that fall outside the white-frequency-noise band and are carried with a model-limited flag; this is one motivation for the random-effect treatment. The full model definitions are given in Appendix C.

### 4.4 Levelling determination of the geopotential difference

The static gravitational-redshift correction is fixed by the geopotential difference between the two clocks’ atomic transition positions, determined by first-order spirit levelling together with simultaneous gravimetry. A levelling line of about 1 113 km was run between the Wuhan benchmark WS-03 (at IAPMST) and the Shanghai benchmark WS-01 (at USTC) through Ezhou, Huangshi, Huanggang, Lu’an, Hefei, Wuhu, Guangde and Suzhou, using digital levels with sight lengths below 30 m and gravity observed at every levelling point with two relative gravimeters. The levelled normal heights were converted to orthometric heights for the geopotential integration,  $\Delta W_{AB} = -\sum_i g_i \Delta H_i$ , where  $g_i$  is the mean gravity and  $\Delta H_i$  the orthometric height difference of the  $i$ -th segment. The resulting geopotential difference between the two atomic transitions is  $\Delta W = 280.042 \pm 0.085 \text{ m}^2 \text{ s}^{-2}$ . The per-kilometre misclosure of the levelling is 0.255 mm, below the first-order limit.

### 4.5 Uncertainty budget

Table 1 lists the uncertainty components of the Yb/Sr ratio. The statistical term is the posterior standard deviation of the Bayesian random-effect model applied to the 16-segment data. The Sr and Yb systematic terms are the published evaluations of the two clocks<sup>21;22</sup>; the value adopted for the Yb clock resolves an internal-versus-published discrepancy in favour of the published evaluation. The static-potential term follows from the levelling uncertainty. The fibre-link and frequency-comb terms are experiment-side upper bounds. The time-varying gravitational-redshift term is carried as an uncertainty item, quantified from the scale over which the segment-averaged shift varies across the campaign, and is not applied as a correction to the reported central value. The components are treated as independent and combined in quadrature.

## A Frequency chain and ratio inversion

The observable is the out-of-loop beat between the transferred 1550 nm carrier and the local Yb-referenced comb. All frequencies in the chain derive from the comb repetition rate  $f_{\text{rep}} = 200$  MHz. The Sr clock’s absolute frequency  $f_{\text{Sr}}$  (698 nm) is related to the comb frequency fed to the 1397 nm branch by

$$f_{\text{comb}}^{1397} = \frac{1}{2} (1 + a) f_{\text{Sr}}, \quad (2)$$

where  $a$  collects the Sr systematic corrections (density, lattice, second-order Zeeman and BBR shifts). The Yb clock’s absolute frequency  $f_{\text{Yb}}$  (578 nm) relates to the 1156 nm comb branch by

$$f_{\text{comb}}^{1156} = \frac{1}{2} (1 + b) f_{\text{Yb}}. \quad (3)$$

The beat is converted to fractional units with the transfer frequency  $F_{1550} = 193.3992$  THz. The static gravitational-redshift correction enters the ratio inversion as a multiplicative factor,

$$(\text{Yb/Sr})_{\text{corr}} = (\text{Yb/Sr})_{\text{raw}} \times (1 + \delta_g), \quad \delta_g = -3.116 \times 10^{-15}, \quad (4)$$

derived from the levelled geopotential difference  $\Delta W = 280.042 \pm 0.085 \text{ m}^2 \text{ s}^{-2}$  (Section 4.4).

## B Data processing and segment definition

The counters sample the beat every second in phase-averaging mode. Samples deviating by more than 10 Hz from the running median are treated as jump points and removed; the retained data are further trimmed at both ends by an endpoint screen that drops points beyond 1% of the peak-to-peak range of the record. The primary data set consists of the longest jump-free record of each accepted segment after this screening. Table 2 lists the per-segment retained counts, ratio deviations and statistical uncertainties; the category column marks the five segments whose stability fits are model-limited. The excluded ninth recorded segment, its parameter provenance and its effect on the combination are described in Sections 4.2 and 2.4.

## C Statistical combination models

**Fixed-effect WLS.** Under the assumption that all segments estimate one common ratio with uncertainties  $u_i$ ,

$$\mu_{\text{WLS}} = \frac{\sum_i w_i R_i}{\sum_i w_i}, \quad w_i = 1/u_i^2, \quad u_{\text{WLS}} = \left( \sum_i w_i \right)^{-1/2},$$

with  $\chi^2 = \sum_i (R_i - \mu_{\text{WLS}})^2 / u_i^2$  and  $\chi_{\text{red}}^2 = \chi^2 / (N - 1)$ . For the primary data set,  $\chi_{\text{red}}^2 = 4.017$  ( $N = 16$ , 15 degrees of freedom,  $p = 2.3 \times 10^{-7}$ ), so the fixed-effect uncertainty under-describes the observed scatter.

**Birge scaling.** The Birge ratio  $B = \sqrt{\chi_{\text{red}}^2} = 2.004$  scales all segment uncertainties by a common factor,  $u_i \rightarrow Bu_i$ ; it assumes the relative sizes of the  $u_i$  are correct but their overall scale is not.

**Mandel–Paule.** An excess scatter  $\xi$  common to all segments is added in quadrature,  $u_{i,\text{eff}}^2 = u_i^2 + \xi^2$ , and  $\xi$  is solved from  $\chi_{\text{red}}^2(\xi) = 1$ . For the primary data set,  $\xi = 2.50 \times 10^{-18}$ .

Table 2: Per-segment data of the primary 16-segment set. Ratio deviations  $y_i = R_i/R_1 - 1$  are relative to the first segment;  $u(y_i)$  is the statistical uncertainty of the segment ratio; “Fit” is *est.* for segments whose stability fit lies in the white-frequency-noise band and *lim.* for model-limited fits.

| Group | $n_{\text{valid}}$ | $y_i$ ( $10^{-18}$ ) | $u(y_i)$ ( $10^{-18}$ ) | Fit  |
|-------|--------------------|----------------------|-------------------------|------|
| 1     | 68 009             | +0.00                | 1.113                   | est. |
| 2     | 29 481             | -3.36                | 1.178                   | lim. |
| 3     | 42 891             | +2.00                | 1.984                   | est. |
| 4     | 64 280             | -4.61                | 0.951                   | est. |
| 5     | 19 532             | -3.38                | 6.040                   | est. |
| 6     | 52 140             | -1.15                | 0.513                   | lim. |
| 7     | 39 526             | +5.31                | 1.173                   | lim. |
| 8     | 43 565             | +0.97                | 4.509                   | est. |
| 10    | 57 938             | -0.76                | 1.230                   | est. |
| 11    | 35 824             | +2.61                | 2.301                   | est. |
| 12    | 40 346             | +0.73                | 1.715                   | lim. |
| 13    | 152 994            | -3.09                | 1.694                   | est. |
| 14    | 68 618             | +1.97                | 1.377                   | est. |
| 15    | 57 649             | -0.25                | 1.293                   | est. |
| 16    | 147 640            | -0.77                | 0.607                   | lim. |
| 17    | 65 983             | -0.89                | 2.142                   | est. |

**Bayesian random effect.** The same observation model,  $R_i \sim \mathcal{N}(\mu, u_i^2 + \xi^2)$ , is treated hierarchically:  $\mu$  and  $\xi$  are both inferred, with flat priors on  $\mu$  and on  $\log \xi$ , and the joint posterior  $p(\mu, \xi \mid \{R_i, u_i\})$  is marginalised numerically over a 40-decade log- $\xi$  grid. The reported central value is the posterior mean of  $\mu$  and the statistical uncertainty is its posterior standard deviation; for the primary data set,  $\xi = 2.83 \times 10^{-18}$ . The excess scatter quantified here also absorbs the correlated component of the time-varying gravitational-redshift term (Section 4.5), which is carried separately as an uncertainty item rather than corrected.

## D Levelling reduction

The orthometric height difference between benchmarks follows from the levelled normal heights after the normal-to-orthometric conversion, with gravity observed at every levelling point. The geopotential difference is integrated segment by segment,  $\Delta W_{AB} = -\sum_i g_i \Delta H_i$ , and combined with the two local offsets between benchmark and clock platform (obtained by trigonometric levelling) and the platform-to-transition offsets from the clock design parameters. The resulting atomic-transition geopotential difference is  $\Delta W = 280.042 \pm 0.085 \text{ m}^2 \text{ s}^{-2}$ , equivalent to an optical fractional shift of  $-3.116 \times 10^{-15}$ .

## E Uncertainty budget details

The components of Table 1 are obtained as follows. The statistical term is the posterior standard deviation of the Bayesian random-effect model (Appendix C). The Sr and Yb terms are the published systematic uncertainties of the two clocks<sup>21;22</sup>; for the Yb clock the published evaluation is adopted. The static-potential term is  $u(\Delta W)/c^2$  from the levelling uncertainty  $0.085 \text{ m}^2 \text{ s}^{-2}$ , giving  $0.946 \times 10^{-18}$ . The fibre-link and comb terms are experiment-side upper bounds of  $1 \times 10^{-19}$  each, carried at  $0.1 \times 10^{-18}$  and  $0.1 \times 10^{-18}$  to preserve the experiment-side margin. The time-varying-redshift term is scaled from the spread of the segment-averaged grav-

itational shift across the campaign and is carried at  $0.5 \times 10^{-18}$ ; it is not applied as a correction. The total standard uncertainty,  $2.12 \times 10^{-18}$ , is the quadrature sum of the seven components.

## Data availability

The frozen, machine-readable result layer underlying this manuscript is archived in the repository (<https://github.com/hjiang555-a11y/gravitation-whsh>) under `results/audit-v1/verified/`: the manifest with SHA-256 15725f5246e87af5a450c8e31ccf0d0ef277d517a493ed1fb44d65a4b86564d4, the derived ledgers and statistics, and the generated audit report. The raw beat files and the levelled geopotential inputs are not redistributed.

## Code availability

The analysis and manuscript pipelines are in the same repository under the Apache License 2.0. The manuscript numbers are generated by `paper/rev2/tools/build_numbers.py`, which reads only the frozen manifest and the curated constants listed in `generated/numbers_manifest.json`.

## References

- [1] K. J. Arnold, M. D. K. Lee, Qi Zhao, Qichen Qin, Zhao Zhang, N. Jayjong, and M. D. Barrett.  $\text{Lu}^+$  optical frequency references with accuracy verified at the 19th digit. *Nature*, 2026. doi: 10.1038/s41586-026-11072-8.
- [2] Bao-lin Zhang, Zi-xiao Ma, Yao Huang, Hui-li Han, Ru-ming Hu, Yu-zhuo Wang, Hua-qing Zhang, Li-yan Tang, Ting-yun Shi, Hua Guan, and Ke-lin Gao. Liquid-nitrogen-cooled  $^{40}\text{Ca}^+$  ion optical clock with a systematic uncertainty of  $4.4 \times 10^{-19}$ . *Phys. Rev. Lett.*, 136: 053202, 2026. doi: 10.1103/vngc-c1xv.
- [3] E. Oelker, R. B. Hutson, C. J. Kennedy, L. Sonderhouse, T. Bothwell, A. Goban, D. Kedar, C. Sanner, J. M. Robinson, G. E. Marti, D. G. Matei, T. Legero, M. Giunta, R. Holzwarth, F. Riehle, U. Sterr, and J. Ye. Demonstration of  $4.8 \times 10^{-17}$  stability at 1 s for two independent optical clocks. *Nature Photonics*, 13(10):714–719, 2019. doi: 10.1038/s41566-019-0493-4.
- [4] Fritz Riehle. Optical clock networks. *Nature Photonics*, 11(1):25–31, 2017. doi: 10.1038/nphoton.2016.235.
- [5] P. Kómár, E. M. Kessler, M. Bishof, L. Jiang, A. S. Sørensen, J. Ye, and M. D. Lukin. A quantum network of clocks. *Nature Physics*, 10(8):582–587, 2014. doi: 10.1038/nphys3000.
- [6] Marco Schioppo, Jochen Kronjäger, Alissa Silva, Rada Ilieva, Chris W. Oates, Jacopo Tunesi, Charles F. A. Baynham, Helen S. Margolis, and Rachel M. Godun. Comparing ultrastable lasers at  $7 \times 10^{-17}$  fractional frequency instability through 2220 km of optical fiber. *Nature Communications*, 13:212, 2022. doi: 10.1038/s41467-021-27884-3.
- [7] Fa-Xi Chen, Li-Bo Li, Jiu-Peng Chen, Kan Zhao, Jian-Yu Guan, Yang Xu, Lei Hou, Fei Zhou, Cheng-Zhi Peng, Qiang Zhang, Hai-Feng Jiang, and Jian-Wei Pan.  $10^{-21}$ -level optical frequency dissemination over 2067 km of noise-loaded field-deployed fiber network. *Light: Science & Applications*, 15:276, 2026. doi: 10.1038/s41377-026-02299-1.
- [8] T. Lindvall, M. Pizzocaro, R. M. Godun, M. Abgrall, others, and H. S. Margolis. Coordinated international comparisons between optical clocks connected via fiber and satellite links. *Optica*, 12(6):843, 2025. doi: 10.1364/optica.561754.

- [9] Marco Pizzocaro, Clara Zyskind, Anne Amy-Klein, Erik Benkler, Sebastien Bize, Davide Calonico, Etienne Cantin, Christian Chardonnet, Cecilia Clivati, Stefano Condio, E. Anne Curtis, Simone Donadello, Sören Dörscher, Chen-Hao Feng, Melina Filzinger, Jacques-Olivier Gaudron, Rachel M. Godun, Irene Goti, Ian R. Hill, Wei Huang, Nils Huntemann, Matthew Johnson, Joshua Klose, Jochen Kronjäger, Alexander Kuhl, Rodolphe Le Targat, Filippo Levi, Burghard Lipphardt, Christian Lisdat, Jerome Lodewyck, Olivier Lopez, Helen S. Margolis, Maxime Mazouth-Lauro, Alberto Mura, Benjamin Pointard, Paul-Eric Pottie, Matias Risaro, Billy I. Robertson, Marco Schioppo, Kilian Stahl, Martin Steinel, Alexandra Tofful, Mads Tø nnes, and Jacob Tunes. International optical clock comparison using the european optical fiber network. *Phys. Rev. Research*, 8:033250, 2026. doi: 10.1103/14bh-ryxs. arXiv:2604.27963.
- [10] N. Dimarcq, M. Gertsvolf, G. Mileti, S. Bize, C. W. Oates, E. Peik, D. Calonico, T. Ido, P. Tavella, et al. Roadmap towards the redefinition of the second. *Metrologia*, 61(1):012001, 2024. doi: 10.1088/1681-7575/ad17d2. arXiv:2307.14141.
- [11] Comité Consultatif du Temps et des Fréquences (CCTF). CCTF strategy 2025–2035. Bureau International des Poids et Mesures, 2025. URL <https://www.bipm.org/documents/20126/2071143/CCTF+Strategy.pdf>.
- [12] Marina Gertsvolf. Task force on the redefinition of the second: Progress report. Comité Consultatif du Temps et des Fréquences (CCTF), BIPM, 2025. URL [https://www.bipm.org/documents/20126/273113451/1\\_CCTF\\_TE\\_28042025\\_Gertsvolf/667adba6-fd36-be37-5237-9be9ccc24d06](https://www.bipm.org/documents/20126/273113451/1_CCTF_TE_28042025_Gertsvolf/667adba6-fd36-be37-5237-9be9ccc24d06).
- [13] Alexander Aepli, Willa J. Arthur-Dworschack, Kyle Beloy, Caitlin M. Berry, Tobias Bothwell, Angela Folz, Tara M. Fortier, Tanner Grogan, Youssef S. Hassan, Zoey Z. Hu, David B. Hume, Benjamin D. Hunt, Kyungtae Kim, Amanda Koepke, Dahyeon Lee, David R. Leibbrandt, Ben Lewis, Andrew D. Ludlow, Mason C. Marshall, Nicholas V. Nardelli, Harikesh Ranganath, Daniel A. Rodriguez Castillo, Jeffrey A. Sherman, Jacob L. Siegel, Suzanne Thornton, William Warfield, and Jun Ye. Atomic clock frequency ratios with fractional uncertainty  $\leq 3.2 \times 10^{-18}$ . *Phys. Rev. Lett.*, 137:033201, 2026. doi: 10.1103/g865-9mk1. arXiv:2512.21428.
- [14] Tara M. Fortier, Andre N. Luiten, and Helen S. Margolis. Optical atomic clocks: defining the future of time and frequency metrology. *Advances in Optics and Photonics*, 2026. in press.
- [15] Jacopo Grotti, Silvio Koller, Stefan Vogt, Sebastian Häfner, Uwe Sterr, Christian Lisdat, Heiner Denker, Christian Voigt, Ludger Timmen, Antoine Rolland, Fred N. Baynes, Helen S. Margolis, Michał Zawada, Ursula Kleinevoss, Nils Huntemann, Ekkehard Peik, Marco Pizzocaro, Simone Donadello, Guglielmo M. Tino, Filippo Levi, Davide Calonico, Alberto Iaco, Marco Tarallo, Elio Rocco, Thomas König, Ali Al-Masoudi, Tanja Mehlstäubler, Christian Lück, Britta Borchert, and Harald Schnatz. Geodesy and metrology with a transportable optical clock. *Nature Physics*, 14:437–441, 2018. doi: 10.1038/s41567-017-0042-3.
- [16] Masao Takamoto, Ichiro Ushijima, Noriaki Ohmae, Toshihiro Yahagi, Kensuke Kokado, Hisa-aki Shinkai, and Hidetoshi Katori. Test of general relativity by a pair of transportable optical lattice clocks. *Nature Photonics*, 14:411–415, 2020. doi: 10.1038/s41566-020-0619-8.
- [17] W. F. McGrew, X. Zhang, R. J. Fasano, S. A. Schäffer, K. Beloy, D. Nicolodi, R. C. Brown, N. Hinkley, G. Milani, M. Schioppo, T. H. Yoon, and A. D. Ludlow. Atomic clock performance enabling geodesy below the centimetre level. *Nature*, 564:87–90, 2018. doi: 10.1038/s41586-018-0738-2. arXiv:1807.11282.

- [18] Andrei Derevianko and Maxim Pospelov. Hunting for topological dark matter with atomic clocks. *Nature Physics*, 10(12):933–936, 2014. doi: 10.1038/nphys3137.
- [19] P. Wcisło, P. Ablewski, K. Beloy, S. Bilicki, M. Bober, R. Brown, R. Fasano, R. Ciurylo, H. Hachisu, K. Hosaka, et al. New bounds on dark matter coupling from a global network of optical atomic clocks. *Science Advances*, 4(12):eaau4869, 2018. doi: 10.1126/sciadv.aau4869.
- [20] Marco Pizzocaro, Mamoru Sekido, Kazuhiro Takefuji, Hideki Ujihara, Hidekazu Hachisu, Nils Nemitz, Masanori Tsutsumi, Tetsuro Kondo, Eiji Kawai, Ryuichi Ichikawa, Kunitaka Namba, Yoshihiro Okamoto, Rumi Takahashi, Junichi Komuro, Cecilia Clivati, Filippo Bregolin, Piero Barbieri, Alberto Mura, Elena Cantoni, Giancarlo Cerretto, Filippo Levi, Giuseppe Maccaferri, Mauro Roma, Claudio Bortolotti, Monia Negusini, Roberto Ricci, Giampaolo Zacchiroli, Juri Roda, Julia Leute, Gérard Petit, Federico Perini, Davide Calonico, and Tetsuya Ido. Intercontinental comparison of optical atomic clocks through very long baseline interferometry. *Nature Physics*, 17:223–227, 2021. doi: 10.1038/s41567-020-01038-6.
- [21] Zhi-Peng Jia, Jie Li, De-Quan Kong, Xiang Zhang, Hai-Wei Yu, Xiao-Yong Liu, Yu-Chen Zhang, Yuan-Bo Wang, Xian-Qing Zhu, Jia-Hao Zhang, Ming-Yi Zhu, Pei-Jun Feng, Xing-Yang Cui, Ping Xu, Xiao Jiang, Xiang-Pei Liu, Peng Liu, Han-Ning Dai, Yu-Ao Chen, and Jian-Wei Pan. Improved systematic evaluation of a strontium optical clock with uncertainty below  $1 \times 10^{-18}$ . *Metrologia*, 63:025002, 2026. doi: 10.1088/1681-7575/ae449e.
- [22] Qiang Zhu, Jingran Shi, Yuechen Zhang, Xiatian Xu, Bing Wang, Dezhi Xiong, Zhuanxian Xiong, Pengcheng Fang, Qunfeng Chen, Lingxiang He, and Baolong Lyu. Ytterbium lattice clock with systematic uncertainty of  $1.3 \times 10^{-18}$  and instability at the  $10^{-19}$  level. *arXiv preprint*, 2026. arXiv:2606.10514; submitted to Metrologia.